

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 313 — Past Questions & Materials
by Renance

Course: PEE 313
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 300

This study pack was compiled and restructured by Renance for PEE 313 students at Federal University of Petroleum Resources, Effurun. It brings together 5 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 5 — PEE 313 Thermodynamics and Phase Behavior

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RE: 0

COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

FIRST SEMESTER EXAMINATION, 2024/2025 session.

PEE 313: THERMODYNAMICS AND PHASE BEHAVIOIR

TIME ALLOWED: 3HRS:00MINS UNIT: 3

ANSWER QUESTION ONE and ANY FOUR QUESTIONS.seccsscsseenseeeeeenes DATE:28/05/2025

QUESTION 1.

0! a. gas is made up of the following components; 35]b of methane, 23]b of ethane and 1.5 1b of propane

i. Express the composition of the gas in weight and mole fractions

ii. | What is the apparent molecular weight of the gas? And What is the gas gravity of the gas?

iii. Calculate the pseudocritical temperature (T_{pc}) and pseudocritical pressure (P_{pc}) of the mixture.

iv. For the gas mixture determine the compressibility factor at a temperature of 150°F and a pressure of 3500psia. Hint;

$$P_{pr} = P/P_{pc}, T_{pr} = T/T_{pe}$$

b. Calculate the pseudo critical properties of the mixture if the gas in question 1(a) also contained impure gases of 31b of hydrogei

sulphide, 10lb of carbon dioxide and 2.51b of nitrogen. pen

Table 1: Composition of an impure Hydrocarbon Sample

Constituents weight Wet rT: Mol lb moles | Mole Pe-psi | Ppepsia | Tpe
fraction CR) | weight fraction(yi)' aia ial

Nitrogen 2.5 28.02

CO, 10

H2S 3 1306

Methane 35 16.04 495.8

Ethane 23

Propane L5 666

Total @15

From wichert & Aziz chart for compositions of H2S and CO2 e= 19

7 ~ , sei Be - $P_{pc}T'_{pe}$

$T_{ipe} = T_{pe} - e$. $P_{pe} = P_{ipe} =$ so nasc-yhasye

e of gas as standard cubic feet.

c.(i) Express the quantity of 101b mol

1a as standard cubic feet.(18) MARKS)

(ii) Express the mass of gas in Ques.

() QUESTION 2.

ai. Draw a typical pressure/volume diagram for a pure component.

ii. Retrograde condensation is generally observed in gas condensate reservoir. Explain.

b.i. What is a phase

ii. In a phase diagram of a multicomponent mixture, within the envelop formed by the

the fluid exists as what? Above the bubble point curve as only what?

iii. State the Dalton's and Raoult's law of Partial Pressure. What is fugacity

c. A reservoir exist at a temperature of 150°F suitable for storing gas. It has an areal size of 5 miles by 2 miles, and is

200ft thick. The average porosity is 20% and there is no water present. How much gas of the composition in Ques 1a can

be stored at a pressure of 3500psia. (1 mile = 5280 ft) (13MARKS)

e bubble-point and dew point curves,

0) QUESTION 3.

(a). Define with the aid of a diagram the following: i

(i) Bubble point (ii) bubble point pressure (i) Dew Point Curve (inh Critical point, B) State the Amagat's law of partial volumes?

How many types of reservoir fluid(s) do we have, name them and briefly explain any two?

(b). The feed to a high pressure separator enters at 10,000 kg/hr and has the following compositions: C_1 ;

80%, C_2 = 10%, C_3 = 7%, C_4 = 3%. the overhead is produced at 2800kg/hr and contains 90% of C_1 , 5% of C_2 , 3% of C_3

and the rest C_4 +. Calculate the weight of the bottom stream and each composition. (13 AMRKS)

4) QUESTION 4.

(a).i Draw a schematic diagram of a 2-stage and 3-stage separation process.

from a two-stage separation of hydrocarbon mixture the table

ij. Considering separator conditions of 76°F and 100 psia j

nt at reservoir conditions of 2,695 psia and 220°F

below. Assume that the mixture is a liquid and its bubble po

= #4

** $T_4 + K_V$

Component Mole Fraction Equilibrium ratio (K_3) $V_e =$

@1A Tpsia n 76 OF

Methane 0.3378 20.8

Ethane 0.0694 4.07

Propane 0.0982 1.16

«_t-butane 0.0133 0.495

n-butane 0.0299 0.343

i-pentane 0.0125 0.142

n-pentane 0.0193 0.108

Hexanes 0.0299 0.0334

| Heavier 0.3897 0.00150

Total 1.0000 as Se eee aintire the tale

Hint: iteration necessary. ey pes

'Calculate the composition and quantities of separator gas and liquid.

ii. Calculate the compositions and quantities of stock tank and liquid, noting that the composition of the feed to the stock

tank is the composition of the liquid from the separator in (i) above, with equilibrium ratio(K_j) of each component given

as 161, 30.7, 8.15, 3.27, 2.30, 0.90, 0.675, 0.20, and 0.0089.) (Hint: using same equation in (i)

iii. Calculate the density of propane plus only.(i3 MARKS)

QUESTION 5.

a. A pure propane is held in a laboratory cell at 100°F and (p) 100psia. Determine the existence state(i.e as a gas or

liquid) of the substance.(hint from your chart)

b.(i) Calculate the vapour pressure(p_v) of propane at 80°F using the Lee and Kesler correlation.

Given: critical temp (T_c) = 666.0, acentric factor (ω)=0.1522, Substance temp(T)=540.

$= 5.92714 - 6.09648 - 1.2886 \ln(T) + 9.16934(7)^\circ$ i -

ha Bok Phas Ty 7 n FL Sag bE

$B = 15.2518 - 15.6875 - 13.4721 \ln(T) + 0.43577)^\circ$

Ss. T;

(if $p < p_y$ the system is entirely what?(iii) If $p > p_y$, the system is entirely what?(iv) If $p = p_y$ the system is entirely

what?(13MARKS) Sirota ast ess ang

©, QUESTION 6.

what you understand by (i) Heat capacity (ii) heat capacity at constant volume (iii) heat capacity at

(a) Define/Explain : I

ermal, isobaric, and isochronic processes.

constant pressure (iv) isoth

(b) A 420 lb.mole hydrocarbon mixture with the component composition and mole fractions given in the table below

undergoes a flash vaporisation and is at equilibrium at 300psia and 200°F. Calculate the mole fraction of gas and liquid of

each component and also, calculate the mole of the both gas and liquid phases in equilibrium.(i3marks)

No°* Component Mole fraction

Methane 0.6203

Ethane 0.2103

Propane 0.0892

i-Butane 0.0215

n-Butane 0.0174 i

i-Pentane 0.0193 :

0.022

n-Pentane

Paper 2 of 5 — PEE 313 THERMODYNAMICS AND PHASE BEHAVIOR 2024|2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

FIRST SEMESTER EXAMINATION, 2024/2025 session.

PEE 313: THERMODYNAMICS AND PHASE BEHAVIOIR

TIME ALLOWED: 3H RS:00MINS UNIT: 3

ANSWER QUESTION ONE and ANY FOUR QUESTIONS.0..0cceeeeeeeeeees DATE:28/05/2025

QUESTION 1,

Ola. gas is made up of the following components; 35lb of methane, 231b of ethane and 1.5 lb of propane.

i. Express the composition of the gas in weight and mole fractions

it. What is the apparent molecular weight of the gas? And What is the gas gravity of the gas?

iii. Calculate the pseudocritical temperature (T_{pc}) and pseudocritical pressure (P_{pc}) of the mixture.

iv. For the gas mixture determine the compressibility factor at a temperature of 150°F and a pressure of 3500psia. Hint;

$$P_{pr} = P/P_{pe}, T_{pr} = T/T_{pe}$$

b. Calculate the pseudo critical properties of the mixture if the gas in question 1(a) also contained impure gases of 3lb of hydroge

sulphide, 10lb of carbon dioxide and 2.5lb of nitrogen. ha

Table 1: Composition of an impure Hydrocarbon Sample

Constituents	weight	Wet	Te	Mol	Ibmoles	Mole	Pecpy	Ppe	psia	Tpe
--------------	--------	-----	----	-----	---------	------	-------	-----	------	-----

fraction (PR) weight fraction(y_i) ae e

Nitrogen fea 28.02 |

CO, 10

HS 3 1306

Methane 35 16.04 495.8

Ethane 23

Propane 1.5 666 B

Total @ 7S |

From wichert & Aziz chart for compositions of H₂S and CO₂ «= 19

ed y ; a af PpcT'pce

pe =Tpe-e. Pipe = P'pe = so yesa—yusye

of 101b mole of gas as standard cubic feet.

c.(i) Express the quantity

1a as standard cubic feet.(18) MARKS)

(ii) Express the mass of gas in Ques.

() QUESTION 2.

ai. Draw a typical pressure/volume diagram for a pure component.

ii. Retrograde condensation is generally observed in gas condensate reservoir. Explain.

b.i. What is a phase

ii. In a phase diagram of a multicomponent mixture, within the envelop formed by the bubble point and dew point curves,

the fluid exists as what? Above the bubble point curve as only what?

iii. State the Dalton's and Raoult's law of Partial Pressure. What is fugacity

c. A reservoir exist at a temperature of 150°F suitable for storing gas. It has an areal size of 5 miles by 2 miles, and is

200ft thick. The average porosity is 20% and there is no water present. How much gas of the composition in Ques 1a can

be stored at a pressure of 3500psia. (1 mile = 5280 ft) (12 MARKS)

() QUESTION 3.

(a). Define with the aid of a diagram the following:

(i) Bubble point (11) bubble point pressure (i) Dew volumes?

How many types of reservoir fluid(s) do we have, name them and briefly explain any two?

(b). The feed to a high pressure separator enters at 10,000 kg/hr and has the following compositions: C₁

80%, C₂ = 10%, C₃ = 7%, C₄ = 3%. the overhead is produced at 2800kg/hr and contains 90% of C₁, 5% of C₂, 3% of C₃

Point Curve (if) Critical point, (8) State the Amagat's law of partial

and the rest C₁ +. Calculate the weight of the bottom stream and each composition.(13 MARKS)

g QUESTION 4.

(a).i Draw a schematic diagram of a 2-stage and 3-stage separation process,

separator conditions of 76°F and 100 psia from a two-stage separation of hydrocarbon mixture the table

ii. Considering separator | ; :

liquid and its bubble point at reservoir conditions of 2,695 psia and 220°F

below. Assume that the mixture is a

one

+S L+ KJ/V

Component Mole Fraction Equilibrium rate (X) V= V= V=

@14 Tpsia and 76 OF

Methane 0.3378 20.8

| Ethane 0.0694 4.07

| Propane 0.0982 1.16

i-butene 0.0133 0.495

P n-butane 0.0299 0.343 L

i-pentane 0.0125 0.142

|_n-pentane 0.093 0.108

Hexanes 0.0299 0.0334

| Heavier 0.3897 0.00150

T Total 1.0000 = . vintare whi

. Hint: iteration necessary. Vo OOS yest

. 5 es e.¢ . .

Calculate the composition and quantities of separator gas and liquid.

ii- Calculate the compositions and quantities of stock tank and liquid, noting that the St Nee of poten if ba stock

tank is the composition of the liquid from the separator in (i) above, with equilibrium ratio(K_j) of eac P given

as 161. 30.7, 8.15, 3.27, 2.30, 0.90, 0.675, 0.20, and 0.0089.) (Hint: using same equation in (i))

iii. Calculate the density of propane plus only. (3 MARKS)

QUESTION 5.

a. A pure propane is held in a laboratory cell at 100°F and (p) 100psia. Determine the existence state(i.e as a gas OF

liquid) of the substance.(hint from your chart)

b.(i) Calculate the vapour pressure(p_y) of propane at 80°F using the Lee and Kesler correlation.

Given; critical temp (T_c) = 666.0, acentric factor (w) = 0.1522, Substance temp(1) = 540.

el day HH

$A = 5.92714 - 6.09648 - | 2886 \ln(7) + 9.16934(7$

$B = 15,2518 - 15.6875 - 13.4721 \ln(T) + 0.4357(T)^\circ$

yf oguehe Ch CEL oe %

T_y

(ii) If $p < p_y$ the system is entirely what? (iii) If $p > p_y$, the system is entirely what? (iv) If $p = p_y$ the system is entirely

what? (3 MARKS) met 4 :

{ } QUESTION 6.

(a) Define/Explain what you understand by (i) Heat capacity (ii) heat capacity at constant volume (iii) heat capacity at

constant pressure (iv) isothermal, isobaric, and isochronic processes.

(b) A 420 lb.mole hydrocarbon mixture with the component composition and mole fractions given in the table below

undergoes a flash vaporisation and is at equilibrium at 300psia and 200°F. Calculate the mole fraction of gas and liquid of

each component and also, calculate the mole of the both gas and liquid phases in equilibrium.(13marks)

No Component Mole fraction

rt | Methane 0.6203

2 Ethane 0.2103

3 Propane 0.0892

4 i-Butane 0,0215

5 n-Butane 0.0174

6 7 1-Pentane 0.0193

7 ~ | n-Pentane 0.022

Paper 3 of 5 — PEE313 thermodynamics & phase behavior 2025

Level 300 • First Semester • PastQuestion

OLEUM RESOURCES

TECHNOLOGY

ENGINEERING

2024/2025 session.

HASE BEHAVOIR

UNEFT: 3

FEDERAL UNIVERSITY OF PETRO

COLLEGE OF ENGINEERING &

DEPARTMENT OF PETROLEUM

FIRST SEMESTER EXAMINATION,

PEE 313: THERMODYNAMICS AND P

TIME ALLOWED: 3HRS:060MINS

ANSWER QUESTION ONE and ANY FOUR QUESTIONS. DATE:28/08/2028

{QUESTION 1.

1a. gas is made up of the following components; 45 lb of methane, 231 lb of ethane and 1.5 lb of propane.

~ . . a . \

position of the gas in weight and mole fractions

i. Express the composition

of the gas? And What is the gas gravity of the gas?

: BP 4 ng } B i 7%

ii, What is the apparent molecular weight

Calculate the pseudocritical temperature (T_{pe}) and pseudocritical pressure (P_{pc}) of the mixture.

iii,

at a temperature of 150°F and a pressure of 3500 psia. Hint,

a ;

iv. For the gas mixture determine the compressibility factor

$P_{pr} = P/P_{pe}$, $T_{pr} = T/T_{pe}$

b. Calculate the pseudo critical properties of the mixture if the gas in question 1(a) also contains

4 lb of pure gases of 31 lb of hydrogen

sulphide, 10 lb of carbon dioxide and 2.5 lb of nitrogen. 4

Table 1; Composition of an impure Hydrocarbon Sample

Constituents weight % Molar weight (lb/lb-mole) | Molar weight (lb/lb-mole) | P_{pe} (psia) T_{pe} (°F)

(: ers fraction (@R) weight fraction(y_i) : : :

[Nitrogen "f25. 28.02 |

CO: 10 a

LBS 3 1306 .

] Methane 35 16.04 495.8 ¢ *

[Ethane 23

Propane 1.5 666

Toval 45 45

From wichert & Aziz chart for compositions of H₂S and CO₂ = 19°

weliees Pee Pye = eee __

Tpe=Tpe-e, Pipe PPC = Ty H₂S(-yH₂S)E

c.(i)Express the quantity of 101b mole of gas as standard cubic feet.

(ii)Express the mass of gas in Ques. 1a as standard cubic feet.(18) MARKS)

QUESTION 2. wt

ai. Draw a typical pressure/volume diagram for a pure component. ~

ii. Retrograde condensation is generally observed in gas condensate reservoir. Explain. “

b.i. What is a phase ~~

ii. In a phase diagram of a multicomponent mixture, within the envelop formed by the bubble point and dew point curves =,

the fluid exists as what? Above the bubble point ee er, @

iii, State the Dalton's and Raoult's law of Partial Pressure. What is fugacity

c, A reservoir exist at a temperature of 150°F suitable for storing gas. It has an areal size of 5 miles by 2 miles, and is

200ft thick. The average porosity is 20% and there is no water present. Figw much gas of the eerrnanitny in Ques 1a can

be stored at a pressure of 3500psia. (1 mile = 5280 ft) (15 MARKS) = Pel rs,

QUESTION 3. PT ite

(a). Define with the aid of a diagram the following: uw Ve .

(i) Bubble point (ii) bubble point pressure (ii) Dew Point Curve (iii) Critical point, (iv) State the Amagat's law of partial

volumes?

How many types of reservoir fluid(s) do we have, name them and briefly explain any two?

(b).The feed to a high pressure separator enters at 10,000 kg/hr and has the following compositions: C:

80%, C₂'= 10%, C₃ = 7%, C_u=3%. the overhead is produced at 2800kg/hr and contains 90% of C_i, oben ae C_s +. Calculate the weight of the bottom steam and each composition.(13 AMRKS)

5% of C_z, 3% Of C₃

(a).i Draw a schematic diagram of a 2-stage and 3-stage S€paration process. \“

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tee from a two-stage separation of hydrocarbon mixture the ta

ii. Consider conditions of 76°F and 100 psia 1 age sepa

be SE eaine® a liquid and its bubble point at reservoir conditions of 2,695 psia and 220°F
SS)

below. :

$L + K_j V$

Equi ite (i = Ve T=

Component Mole Frection smd seh OF Vv

0.3378 20.8 _

a : 0.0694 4.07 _

Propane 0.0982 1.16 7 7

i-butane 0.0133 0.495 _

n-butane 0.0299 0.343

i-pentane 0.0125 0.142 ,

A-pentane 0.0193 0.108 y

Hexanes 0.0299 0.0334

Heavier 0.3897 0.00150

Total 1.0000 ee re , Atenas 2 '

.. . _ Hint: iteration necessary, _

Calculate the composition and quantities of separator gas and liquid.

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ii. Calculate the compositions and quantities of stock tank and liquid, noting that the composition of the feed to the stock

tank is the composition of the liquid from the separator in (i) above, with equilibrium ratio(K_j) of each component given

as 161, 30.7, 8.15, 3.27, 2.30, 0.90, 0.675, 0.20, and 0.0089.) (Hint: using same equation in (i))

iii. Calculate the density of propane plus only. (3 MARKS)

QUESTION 5.

a. A pure propane is held in a laboratory cell at 100°F and (p) 100psia. Determine the existence state(i.e as a gas or

liquid) of the substance.(hint from your chart)

b.(i) Calculate the vapour pressure(p_v) of propane at 80°F using the Lee and Kesler correlation.

Given: critical temp (T_c) = 666.0, acentric factor (ω) = 0.1522, Substance temp(T) = 540.

Pts Tet aa Ep op

$A = 5.92714 - 6.09648 - 1.2886 \ln(T) + 0.16934(7 \text{ Ser! SLEEPS})$

$B = 15.2518 - 15.6875 \sim 13.4721 \ln(T) + 0.4357(7)^\circ$

hea As Pr the ci é Parr what?(iii) If $p > p_y$ the system is entirely what?(iv) If $P = P_y$ the system is entirely

QUESTION 6. * oe

(a) Define/Explain what you understand by (i) Heat ϕ i

Sats s pacity (ii) heat capacity at constant ; iii i

constant pressure (iv) isothermal, isobaric, and isochronic processes. nt volume (iii) heat capacity at

Mole fraction

0.6203

0.2103

0,0892

0.0215

0.0174

0.0193

IO Q00 0 -

9.022

Paper 4 of 5 — PEE 313 Thermodynamics and PVT behaviour

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF ENGINEERING & TECHNOLOGY.

DEPARTMENT OF PETROLEUM ENGINEERING

FIRST SEMESTER EXAMINATION, 2023/2024

PEE 313: THERMODYNAMICS AND PVT PHASE BEHAVOIR

TIME ALLOWED: 3HRS:00MIN (UNIT: 3) ANSWER ANY 5 QUESTIONS...

QUESTION 1.(a) Write short notes on the following: (i) Pressure-volume di

is generally observed in gas condensate reservoir. Explain.

(bp. i. In a phase diagram of a multicomponent mixture, within the eny

€ fluid exists as what? Above the bubble point curve as only what?

li. State the Dalton's law of Partial Pressure.

DATE:23/05/2024

Actor. (iii)Retrograde condens

agram. (ii)

ation

elop formed by the bubble point and dew point curves,

(c). Define the following: (i)Phase Gi) Dew Point Curve (iii) Critical point (iv)

State the Amagat's law of partial volumes?

(v) How many types of reservoir fluid(s) do we have, name them and briefly

explain any two?(15 MARKS)

QUESTION 2. (a) Draw a schematic diagram of a 2-stage and 3-stage separation process.

(b) Considering separator conditions of 76°F and 100

psia from a two-stage separation of hydrocarbon mixture below. Assume

« that the mixture isa liquid and its bubbie point at reser

voir conditions of 2,695 psia and 220°F

L+ KV

Component Mole Fraction | Eauilibrium ratio V= V= V=

(kj)

@14.7psia n 76 oF | |

Methane 0.3378 20.8 | ah

Ethane 0.0694 4.07 | “|

Propane 0.0982 1.16 | +

i-butane 0.0133 0.495 : | a

i n-butane 0.0299 0.343 | |

i-pentane 0.0125 0.142 | a

7 | H-pentane | _ 0.0193 i a nn aca wii ee mica

Hexanes | _ 0.0334

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Hint: iteration necess

OOS eae ae ees!

alculate the composition and quantities of separator gas and liquid.

ii. Calculate the compositions and quantities of stock tank and liquid, noting that the composition of the feed to the stock tank

is the composition of the liquid from the separator in (i) above, with equilibrium ratio(sj) of each component given as 161,

30.7, 8.15, 3.27, 2.30, 0.90, 0.675, 0.20, and 0.0089.) (Hint: using same equation in (i))

iii. Calculate the density and molecular weight of the stock tank oil. (15 MARKS)

QUESTION 3. (a) gas cylinder contains methane at 1000 psia and 70oF, if the cylinder has a yolume of 3 cu.ft assuming

isan ge

QUESTION 4. Calculate the pseudo critical properties of the mixture if the gas in Question 3 (b) also contained impure gases

Of 3tb of hydrogen sulphide, 101b of carbon dioxide and 2.5lb of nitrogen.

‘Table 1: Composition of an impure Hydrocarbon Sample

| Constituents | weight | Wet a

Mol lb Mole Pe- | Poe

13 fraction | {OR | weight | moles fraction(yi) | psi_ | psia

| Nitrogen ps O55 | 227 |e. 0/0891 6 98 ff le NG 331 5.3.44

coz 10 + 22%} 543 | yf tl lp-927 9. BAG didi Uedodh 99.4)

H2S 3 f- 6611673 [Sy lp ORK go 8 bap 25-5] 16 15

Methane 25 6 666 344 [ie Issel o- d tt la dong Hy. 3ç4

Ethane 310 -Gç4550_ |9).o?ln =999 9. > 65/68 ROM tad. C5

Propane 1.5 Ore 033] 666 [Hy 04/)-os4] Ao GS-1 6-5eh Bd oa

Total 45 i

125° GI GEE). 224

From wichert & Azis chart for compositions of H₂S and CO, =19

$T_{pe} = T_{pe-e}$, $P_{pe} = P_{pc} = P_{pcT}$ pe g «1 4

$T_{pc} + y_{H_2S}(1 - y_{H_2S})$ a 1D; §

Express the quantity of 11b mole of gas as standard cubic feet. (15 MARKS)

QUESTION 5. (a). Besides the y

Petroleum Industry. For each E

= b8K°3 24-19

an der Waals Equation of State, discuss four(4) Equations of State commonly used in the
quation of State, ensure that you highlight: <= ob< \$2 q NE »

i. The assumptions the authors made in dey eloping the equation of state; r n —

ii. The eqvation(s): iii.The applications of the Equation of State in the Petroleum Industry. } -f Fe b

iv. The limitations of the Equation of State. aa

b. A pure propane is held in a labor atory cell at 80 OF and 200psia. Determine the “existence
state”+fi.e. as a pas or liquid) of

the substance. ' Hy §

} -W ess) »

ce. Calculate the vapour pressure of propane at 80 OF using the Lee and Kesler correlation.) Ye

Given: critical temp (T_c) =666.0, acentric factor (z)=0.1522, Substance temp(T)=540. :

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4, 1 sf

5 | N-Butan VEIK (0135 20

G. | i-Pentane (e713

7 | N-Pentane z S1€ “14S ¢

Paper 5 of 5 — PEE 313

Level 300 • First Semester • Handout

PETE 310

Review Lecture # 7

Three & Multicomponent Mixtures...

Plus

Lecture # 8 - Chapter 5

Ternary Diagrams: Review

L

.1 .9

.2 .8

.3 .7

.4 .6

.5 .5

.6 .4

.7 .3

.8 .2

.9 .1

1 0

H0 .1 .2 .3 .4 .5 .6 .7 .8 .9 1 I

Ternary Diagrams: Review

Pressure Effect

C1 C1 C1

Gas

2-phase 2-phase

Liquid Liquid

nC5 C3 nC5 C3 nC5 C3

p=14.7 psia p=380 psia p=500 psia

C1 C1 C1

2-phase

2-phase

Liquid Liquid

Liquid

nC5 nC5 nC5 C3

p=1500 psia C3 p=2000 psia C3 p=2350 psia

Ternary Diagrams: Review

C1

Dilution Lines .1 .9

.2 .8

.3 .7

.4 .6

.5 .5

.6 .4

.7 .3

.8 .2

.9 x .1

1 0

C10 0 .1 .2 .3 .4 .5 .6 .7 .8 .9 1 n-C4

Ternary Diagrams: Review

Quantitative Representation of Phase

Equilibria - Tie (or equilibrium) lines

Tie lines join equilibrium conditions of the gas and the liquid at a given pressure and temperature

Bubble point curve gives the liquid composition

Dew point curve gives the gas composition

Ternary Diagrams: Review

Quantitative Representation of Phase

Equilibria - Tie (or equilibrium) lines

All mixtures whose overall composition (z_i) is along a tie line have the SAME equilibrium gas (y_i) and liquid composition (x_i), but the relative amounts on a molar basis of gas and liquid (f_v and f_l) change linearly (0 – vapor at B.P., 1 – liquid at B.P.).

Illustration of Phase Envelope and

Tie Lines

C 1

.1 .9

.2 .8

.3 .7

.4 .6

.5 .5

CP

.6 .4

.7 .3

.8 .2

.9 .1

1 0

C10 0 .1 .2 .3 .4 .5 .6 .7 .8 .9 1 n-C4

Uses of Ternary Diagrams

Representation of Multi-Component Phase

Behavior with a Pseudoternary Diagram

Ternary diagrams may approximate phase behavior of multi-component mixtures by grouping them into 3 pseudocomponents

heavy (C7+)

intermediate (C2-C6)

light (C1, CO₂ , N₂- C1, CO₂-C2, ...)

Uses of Ternary Diagrams

Miscible Recovery Processes

C1

.1 .9

.3

.2 .8

.7

Solvent2

.4 .6

.5 .5

.6 .4

.7 Solvent1 A

.3

.8 .2

.9 0 .1

C7+

1

0 .1 .2 .3 .4

oil

.5 .6 .7 .8 .9

0

1 C2-C6

Exercise

Find overall composition of mixture made with 100

moles oil "O" + 10 moles of mixture "A".

_____ C1

_____ .2

.1 .9

.8

_____ .3 .7

_____ .5

.4 .6

.5

_____ .6 .4

A

_____ .8

.7 .3

.2

.9 O .1

1 0

C7+ 0 .1 .2 .3 .4 .5 .6 .7 .8 .9 1 C2-C6

Mixture C1 -nC4 -C10

liquid phase @ p=2500 psi, T=160 oF

x_{C1} x_{C4} x_{C10} x y

0.494 0.000 0.506 0.247 0.427816

C1

0.496 0.101 0.403 0.349 0.429548

0.509 0.197 0.294 0.4515 0.440807 liquid phase

gas phase

0.534 0.280 0.186 0.547 0.462457

gridlines

0.600 0.320 0.080 0.62 0.519615

triangle

0.670 0.310 0.020 0.645 0.580237

equilibrium

gas phase

y_{C1} y_{C4} y_{C10}

0.994 0.000 0.006 0.497 0.860829

0.965 0.028 0.007 0.5105 0.835714

0.927 0.064 0.009 0.5275 0.802805

0.874 0.115 0.011 0.552 0.756906

0.800 0.186 0.014 0.586 0.69282

0.740 0.247 0.013 0.617 0.640859

C10 nC4

0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1

http://www.pe.tamu.edu/barrufet/public_html/PETE310/index.html

Practice Ternary Diagrams

Pressure Effect

T=180F Pressure Effect T=180F Pressure Effect

P=14.7 psia P=200 psia

C1-C3-C10

O O

T=180F Pressure Effect T=180F Pressure Effect

P=400 psia P=600 psia

O O

Practice Ternary Diagrams

Pressure Effect

T=180F Pressure Effect T=180F Pressure Effect

P=1000 psia P=1500 psia

O O

T=180F T=180F T=180F

P=2000 psia P=3000 psia P=4000 psia

O O O

Practice Ternary Diagrams

Temperature Effect

T=100F Temperature Effect T=150F Temperature Effect

P=2000 psia P=2000 psia

O O

T=200F Temperature Effect T=300F Temperature Effect

P=2000 psia P=2000 psia

O O

Practice Ternary Diagrams

Temperature Effect

T=350F Temperature Effect T=400F Temperature Effect

P=2000 psia P=2000 psia

O O

T=450F Temperature Effect

P=2000 psia

O

Pressure-Temperature Diagram

for Multicomponent Systems

1-Phase 1-Phase

CP

Reservoir Pressure

60%

0%

20%

2-Phase

Reservoir Temperature

Changes During Production and
Injection

t

1 Production

t

2

Pressure

Gas

Injection

t

3

Temperature

PETE 310

Lecture # 8: Five Reservoir Fluids
(Chapter 5)

Pressure vs. Temperature
Diagrams

Used to visualize the fluids production
path from the reservoir to the surface

To classify reservoir fluids

Visualize miscible processes

Pressure-Temperature Diagram
for Multicomponent Systems

1-Phase 1-Phase

Reservoir Pressure CP

60%

0%

20%

2-Phase

Reservoir Temperature

Why do we need to classify
Reservoir Fluids?
Determine fluid sampling
Determine types and sizes of surface equipment
Dictate depletion strategy
Determine selection of EOR method
Determine techniques to predict oil & gas
reserves
Determine Material Balance calculations
Phase Envelopes
Cricondenbar
Bubblepoint Curve
Critical
Point Fixed
Composition
Dew Point Curve
Quality
Pressure
75% Lines
50%
Cricondentherm
25%
Temperature
Classification of Reservoirs based
on Phase Diagram
Gas Reservoirs (Single Phase)
Gas Condensate Reservoirs (Dew-Point
Reservoirs):
Undersaturated Solution-Gas Reservoirs
(Bubble-Point Reservoirs):
Phase Diagram of a
Dry Gas Reservoir
Initial Reservoir
Conditions
CP
Pressure
Path of Production
Separator Conditions
Temperature

Phase Diagram of a
Wet Gas Reservoir

CP Initial Reservoir
Conditions

Pressure

Path of Production

Separator Conditions

Temperature

Phase Diagram of a
Retrograde Gas Reservoir

Initial Reservoir

Conditions

CP

Pressure

Path of Production

Separator Conditions

Temperature

Phase Diagram of a
Volatile Oil Reservoir

Initial Reservoir

Conditions

CP

Path of Production

Pressure

Separator Conditions

75%

25%

50%

Temperature

Phase Diagram of a
Black Oil Reservoir

Initial Reservoir

Conditions

Path of Production

CP

Pressure

75% 50%

25%

Separator Conditions

Temperature

Phase envelopes of different mixtures
with different proportions of same HC
components

7000

TR Critical Points

6000

Volatile I

5000

Condensate Volatile II

Pressure (psia)

4000

3000

2000 Wet Gas

Black Oil

1000

Dry Gas

0

-200 -100 0 100 200 300 400 500 600 700 800

o

Temperature F

Typical Reservoir Fluid Compositions

Component Black Oil Volatile Oil Gas Condensate Wet Gas Dry Gas

C1 48.83 64.36 87.07 95.85 86.67

C2 2.75 7.52 4.39 2.67 7.77

C3 1.93 4.74 2.29 0.34 2.95

C4 1.60 4.12 1.74 0.52 1.73

C5 1.15 3.97 0.83 0.08 0.88

C6 1.59 3.38 0.60 0.12

C7 42.15 11.91 3.80 0.42

MwC7 225 181 112 157

GOR 625 2000 18,200 105,000 -

o

Tank API 34.3 50.1 60.8 54.7 -

Liquid Greenish Medium Light Water -

Color Black Orange Straw White

Compositional

Distribution
of
Reservoir
Fluids

Classification of Reservoirs
based on Production and PVT
data

GAS CONDENSATE RESERVOIRS:

GOR between 70,000-100,000 SCF/STB

Density greater than 60 °API

Light in color

C7+ composition < 12.5%

Classification of Reservoirs
based on Production and PVT
data

VOLATILE OIL RESERVOIRS:

GOR between 1,000-8,000 SCF/STB

Density between 45-60 °API

Oil FVF greater than 2.00 (high
shrinkage oils)

Light brown to green in color

C7+ composition > 12.5%

Classification of Reservoirs based
on Production and PVT data

BLACK OIL RESERVOIRS:

GOR less than 1,000 SCF/STB

Density less than 45 °API

Reservoir temperatures less than 250 °F

Oil FVF less than 2.00 (low shrinkage oils)

Dark green to black in color

C7+ composition > 30%

Assignment

Read and make a summary of revised &
newer criteria for classification of
Reservoir Fluids from given paper by
William D. McCain in JPT September 1994

JPT paper Study Guide

What are the distinctive features of black
oils in terms of

Initial GOR & GOR vs time

Initial API & API vs time

Compositions

Color

JPT paper Study Guide

What are the distinctive features of
volatile oils in terms of

Initial GOR & GOR vs time

Initial API & API vs time

Compositions

Color

JPT paper Study Guide

What are the distinctive features of
Condensate gases in terms of

Initial GOR & GOR vs time

Initial API & API vs time

Compositions

Color

JPT paper Study Guide

What are the distinctive features of Dry
gases in terms of

Initial GOR & GOR vs time

Compositions

PETE 310

Lecture # 12

Properties of Dry Gases

(Chapter 6 - pages 165-178)

Phase Diagram of a

Dry Gas Reservoir

Initial Reservoir

Conditions

CP

Pressure

Path of Production

Separator Conditions

Temperature

Dry Gas Reservoir Characteristics

No hydrocarbon liquids produced at
surface or minimal production (or $GOR >$

100,000 SCF/STB)

May produce water

Mostly methane

Standard Conditions

Unify volumes to common grounds for
sales and regulatory purposes

$T = 60\text{ }^{\circ}\text{F}$

$P = 14.65 - 15.025$ (State dependent see
Table 6-1)

Practice evaluating VM For TX

$VM = RT_{sc}/P_{sc}$

More Practice Problems

A dry gas well with a $SG = 0.56$ in Aggie-
field is producing 70,000 SCF/day

How many lb-moles are being produced
daily?

How many pounds?

If the gas is made of just C_1 and C_2 , what is
the gas composition?

Reservoir Engineering Properties of
Dry Gases

Gas formation volume factor B_g

Reservoir Conditions

Standard Conditions

Gas Formation Volume Factor

[res bbl/SCF] or [ft³/SCF]

Volume of an arbitrary amount
of gas at reservoir T & P

Volume of SAME amount at
standard T & P

V_R

$B_g \square$

V_{SC}

Gas Formation Volume Factor

[res bbl/SCF] or [ft³/SCF]

$Z nRT$

$B_g \square P$

$Z_{SC} nRT_{SC}$

P_{SC}

Gas Formation Volume Factor

Gas Formation Volume Factor

[res bbl/SCF] or [ft³/SCF]

Gas Formation Volume Factor

B_g

Pressure

Example of Use of Dry Gas

Properties in Material Balance

Equation

Much more of these in PETE 323

Classical Reservoir Engineering Analysis

Volumetric

Isomaps

Structural maps

Geological wisdom

Material Balance

Production data

PVT data

Fluid Displacement

Fractional Flow theory

Waterflooding

Limitations of Material Balance Methods

No spatial distribution of pressures and saturations

No Indication of Unswept or Uncontacted Areas

Does Not Use Spatial Information

Reservoir properties are average properties

Reservoir is a tank

Processes Included in

GMBE

GENERAL MATERIAL BALANCE EQUATION

(GMBE)

Gas

Injection OIL+GAS+WATER

PRODUCTION

Gas

Oil

Water

Gas

Oil Approximates reservoir

as a tank with given

Water boundaries and

uniform T and P.

Drive

Water MB Calculation is a

very gross

Water approximation of the

reservoir performance,

Influx but one that is very

useful in practice

Sequence of Stirred Tanks Model

Sources of Reservoir Energy

Liquid and rock compressibility,

Gas in solution,

Gas in the gas cap,

Water influx,

Gravity Segregation

Volumetric gas material

balance

Moles of gas

$$n_i = p_i V / z_i R T \quad n_2 = p_2 V / z_2 R T$$

$$V = X \quad V = X$$

Original pressure at New pressure at
reservoir temperature reservoir temperature

Time 1 Time 2

A Simple Mass Balance

$$\Delta M$$

P

Beginning End

MB Time Interval ME

$$\Delta M$$

I

$$\text{Mass Balance: } M_E - M = \Delta M_I - \Delta M$$

B P

On a volumetric basis

$$V_E - V = V - V$$

B I P

Volumetric gas material

balance

Moles of gas produced = $n_1 - n_2$

380.7 scf per lb mol of gas in Texas (@60 deg F and 14.65 psia)

Volume of gas produced = $380.7 \cdot (n_1 - n_2)$ scf

Material Balance Derivation

$p_i V = n_i z_i RT$

Initial state

before production

After some

Production

$p V = n_i z_i RT$

Moles left in reservoir

Ratio Equations and rearrange

$p V = n_i z_i RT$

$p V = n_i z_i RT$

$p V = n_i z_i RT$

$p V = n_i z_i RT$

$p V = n_i z_i RT$

$p V = n_i z_i RT$

Material Balance Derivation

$p V = n_i z_i RT$

$p V = n_i z_i RT$

$p V = n_i z_i RT$

Express moles produced and initial in terms of standard volumes

$p_{sc} G_p = p_{sc} G_i$

$n_p = n_i$

$R T_{sc} = R T_{sc}$

P/z plot Volumetric reservoir graphic

interpretation--

$p V = n_i z_i RT$

$p V = n_i z_i RT$

$p V = n_i z_i RT$

P/z@original pressure

P/z

$G_p = G_i @ P=0$

G_p

Material Balance Exercise

Volumetric Depletion of a Dry Gas

Reservoir

Gp

Pressure Gp (SCF) Z (MMSCF) P/Z

6000 0 1.0135 0 5920.078934

5300 7737328.244 1.0006 7.7373282 5296.821907

4520 13536207.06 0.9627 13.536207 4695.128285

3640 23488811.39 0.8972 23.488811 4057.066429

3100 30905455.55 0.863 30.905456 3592.12051

2600 38671313.9 0.8502 38.671314 3058.103976

2120 47318522.34 0.8525 47.318522 2486.803519

2028 49122795.64 0.855 49.122796 2371.929825

1760 53512916.85 0.8617 53.512917 2042.474179

1570 58302139.98 0.8765 58.30214 1791.21506

Material Balance Exercise

Volumetric Gas Depletion

7000

$$y = -72.934x + 5920.1$$

6000

$$R^2 = 0.9942$$

5000

4000

P/z

3000

$$2000 \text{ p}_{\text{pi}} - G \text{ p}_{\text{p}}$$

$$= 1000 \text{ z}_{\text{zi}} - G$$

$$1000 \text{ z}_{\text{zi}} - G$$

0

0.00 10.00 20.00 30.00 40.00 50.00 60.00 70.00

Gp (MMSCF)

Related Possible Exam Questions

Find the gas in place

Find recoverable gas at a certain
abandonment pressure

Determine initial reservoir pressure

Determine reservoir pore volume

More Gas Properties

Viscosity

Heat of combustion

Introduction to

Thermodynamics

Compiled by Dr. Mukhtar Fatihu Hamza

Thermodynamics Lecture Note by Dr.

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Thermodynamics

- Thermodynamics is the study of the effects of work, heat, and energy on a system
- Thermodynamics is only concerned with macroscopic (large-scale) changes and observations
- Thermodynamics deals with stability of systems. It tells us 'what should happen?'. 'Will it actually happen(?)' is not the domain of thermodynamics and falls under the realm of kinetics.

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Thermodynamics (TD): perhaps the most basic science

□ One branch of knowledge that all engineers and scientists must have a grasp of is thermodynamics.

□ In some sense thermodynamics is perhaps the 'most abstract subject' and a student can often find it very confusing if not 'motivated' strongly enough.

□ Thermodynamics can be considered as a 'system level' science- i.e. it deals with descriptions of the whole system and not with interactions at the level of individual particles.

I.e. it deals with quantities (like T,P) averaged over a large collection of entities (like molecules, atoms).

This implies that questions like: "What is the temperature or entropy of an atom?"; do not make sense in the context of thermodynamics.

□ TD puts before us some fundamental laws which are universal in nature (and hence applicable to fields across

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disciplines).

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The language of TD

□ To understand the laws of thermodynamics and how they work, first we need to get the terminology right. Some of the terms may

look familiar (as they are used in everyday language as well)- but their meanings are more 'technical' and 'precise', when used in TD and hence we should not use them 'casually'.

□ System is region where we focus our attention (Au block in figure).

□ Surrounding is the rest of the universe (the water bath at constant 'temperature').

□ Universe = System + Surrounding (the part that is within the dotted line box in the figure below)

□ More practically, we can consider the 'Surrounding' as the immediate neighbourhood of the system (the part of the universe at large, with which the system 'effectively' interacts).

In this scheme of things we can visualize: a system, the

In TD we usually do not worry about surrounding and the universe at large.

the universe at large!

□ Things that matter for the surrounding: (i) T, (ii) P, (iii) ability to: do work, transfer heat, transfer matter, etc. Parameters for the system: (i) Internal energy, (ii) Enthalpy, (iii) T, (iv) P, (v) mass, etc. Thermodynamics Lecture Note by Dr.

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Open, closed and isolated systems

□ To a thermodynamic system two 'things' may be added/removed:

□ energy (in the form of heat &/or work) □ matter.

□ An open system is one to which you can add/remove matter (e.g. a open beaker to which we can add water). When you add matter- you also end up adding heat (which is contained in that matter).

□ A system to which you cannot add matter is called closed.

Though you cannot add/remove matter to a closed system, you can still add/remove heat (you can cool a closed water bottle in fridge).

□ A system to which neither matter nor heat can be added/removed is called isolated.

A closed vacuum 'thermos' flask can be considered as isolated.

Type of boundary Interactions Mass

Open All interactions possible (Mass, Work, Heat)

Closed Matter cannot enter or leave

Interactions possible

Semi-permeable Only certain species can enter or leave

Work

Insulated Heat cannot enter or leave

Rigid Mechanical work cannot be done*

Heat

Isolated No interactions are possible**

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☐ Matter is easy to understand and includes atoms, ions, electrons, etc.

☐ Energy may be transferred ('added') to the system as heat, electromagnetic radiation etc.

☐ In TD the two modes of transfer of energy to the system considered are Heat and Work.

☐☐ Heat and work are modes of transfer of energy and not 'energy' itself.

☐☐ Once inside the system, the part which came via work and the part which came via heat, cannot be distinguished*. More sooner on this!

☐☐ Before the start of the process and after the process is completed, the terms heat and work are not relevant.

☐ From the above it is clear that, bodies contain internal energy and not heat (nor work!).

☐ Matter when added to a system brings along with it some energy. The 'energy density' (energy per unit mass or energy per unit volume) in the incoming matter may be higher or lower than the matter already present in the system. Thermodynamics Lecture Note by Dr.

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Processes in TD We will deal with some of them in detail later on

☐ Here is a brief listing of a few kinds of processes, which we will encounter in TD:

☐ Isothermal process → the process takes place at constant temperature (e.g. freezing of water to ice at -10°C)

☐ Isobaric → constant pressure

(e.g. heating of water in open air → under atmospheric pressure)

☐ Isochoric → constant volume

(e.g. heating of gas in a sealed metal container)

☐ Reversible process → the system is close to equilibrium at all times (and infinitesimal alteration of the conditions can restore the universe (system + surrounding) to the original state. (Hence, there are no truly reversible processes in nature).

☐ Cyclic process → the final and initial state are the same. However, q and w need not be zero.

☐ Adiabatic process → dq is zero during the process (no heat is added/removed to/from the system)

☐ 9/27/2021

A combination of the above are also possible: e.g. 'reversible adiabatic process'.

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Temperature

□ Though we all have a feel for temperature ('like when we are feeling hot'); in the context of TD temperature is technical term with 'deep meaning'.

□ As we know (from a commons sense perspective) that temperature is a measure of the 'intensity of

heat'. 'Heat flows' (energy is transferred as heat) from a body at higher temperature to one at lower

temperature. (Like pressure is a measure of the intensity of 'force applied by matter' → matter (for now a fluid) flows from region of higher pressure to lower pressure).

□ That implies (to reiterate the obvious!) if I connect two bodies □ (A)-one weighing 100kg at 10°C and the other (B) weighing 1 kg at 500°C, then the 'heat will flow' from the hotter body to the colder body (i.e. the weight or volume of the body does not matter).

□ But, temperature comes in two important 'technical' contexts in TD:

1 □ it is a measure of the average kinetic energy (or velocity) of the constituent entities (say molecules)

2 □ it is the parameter which determines the distribution of species (say molecules) across various energy states available.

Heat flow

A direction B

500°C

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10°C 8

Temperature as a parameter determining the distribution of species across energy levels

□ Let us consider various energy levels available for molecules in a system to be promoted to.

□ At low temperatures the lower energy levels are expected to be populated more, as compared to higher energy levels.

As we heat the system, more and more molecules will be promoted to higher energy levels.

□ The distribution of molecules across these energy levels is given by:

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Few points about temperature scales and their properties

□ Celsius (Fahrenheit, etc.) are relative scales of temperature and zero of these scales do not have a fundamental significance. Kelvin scale is a absolute scale.

Zero Kelvin and temperatures below that are not obtainable in the classical sense.

□ Classically, at 0K a perfect crystalline system has zero entropy (i.e. system attains its minimum entropy state). However, in some cases there could be some residual entropy due to degeneracy of states (this requires a statistical view point of entropy).

□ At 0K the kinetic energy of the system is not zero. There exists some zero point energy.}

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Pressure

□ Pressure* is force per unit area (usually exerted by a fluid on a wall**).

□ It is the momentum transferred (say on a flat wall by molecules of a gas) per unit area, per unit time. (In the case of gas molecules it is the average momentum transferred per unit area per unit time on to the flat wall).

□ $P = \text{momentum transferred/area/time}$.

□ Pressure is related to momentum, while temperature is related to kinetic energy.

Wall of a container

‘Crude schematic’

of particles

impinging on a

wall.

* ‘Normal’ pressure is also referred to as hydrostatic pressure.

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** Other agents causing pressure could be radiation, macroscopic objects impinging on a wall, etc.

Example

A tank has a volume of 0.5 m³ and contains 10 kg of an ideal gas having a molecular weight of 24. The temperature is 25 °C. What is the pressure?

The gas constant is determined first:

We now solve for P:

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Heat and Work

□ Work (W) in mechanics is displacement (d) against a resisting force (F). $W = F \cdot d$

□ Work has units of energy (Joule, J).

□ Work can be expansion work ($P\Delta V$), electrical work, magnetic work etc. (many sets of stimuli and their responses).

□ Heat as used in TD is a tricky term (yes, it is a very technical term as used in TD).

□ The transfer of energy as a result of a temperature difference is called heat.

□ “In TD heat is NOT an entity or even a form of energy; heat is a mode of transfer of

energy" [1].

- "Heat is the transfer of energy by virtue of a temperature difference" [1].
- "Heat is the name of a process, not the name of an entity" [1].
- "Bodies contain internal energy (U) and not heat" [2].
- The 'flow' of energy down a temperature gradient can be treated mathematically by considering heat as a mass-less fluid [1] → this does not make heat a fluid!

Expansion work

To give an example (inspired by [1]):

assume that you start a rumour that there is 'lot of' gold under the class room floor. This rumour 'may' spread when persons talk to each other.

The 'spread of rumor' with time may be treated mathematically by equations, which have a form similar to the diffusion equations (or heat

transfer equations). This does not make 'rumour' a fluid!

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[1] Four Laws that Drive the Universe, Peter Atkins, Oxford University Press, Oxford, 2007. [2] Physical Chemistry, Ira N Levine, Tata McGraw Hill Education Pvt. Ltd., New York (2002).

- Work is coordinated flow of matter.
 - Lowering of a weight can do work
 - Motion of piston can do work
 - Flow of electrons in conductor can do work.
 - Heat involves random motion of matter (or the constituent entities of matter).
 - Like gas molecules in a gas cylinder
 - Water molecules in a cup of water
 - Atoms vibrating in a block of Cu.
 - Energy may enter the system as heat or work.
 - Once inside the system:
 - it does not matter how the energy entered the system* (i.e. work and heat are terms associated with the surrounding and once inside the system there is no 'memory' of how the input was received and
 - the energy is stored as potential energy (PE) and kinetic energy (KE).
 - This energy can be withdrawn as work or heat from the system.
- * As Aktins put it: "money may enter a bank as cheque or cash□ but once inside the bank there is no difference".

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Example

Consider as a system the gas in the cylinder in Fig. 4.2; the cylinder is fitted with a piston on which a number of small weights are placed. The initial pressure is 200 kPa and the initial volume of the gas is 0.04 m³.

1. Let a Bunsen burner be placed under the

cylinder, and let the volume of the gas increase to 0.1 m³ while the pressure remains constant. Calculate the work done by the system during the process.

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2. Consider the same system and initial conditions, but at the same time the Bunsen burner is under the cylinder and the piston is rising, let weights be removed from the piston at such a rate that, during the process, the temperature of the gas remains constant. If we assume that the ideal-gas model is valid, and that is a polytropic process with exponent $n = 1$. Therefore,

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Reversible process 'Reversible' is a technical term (like many others) in the context of TD.

□ A reversible process is one where an infinitesimal change in the conditions of the surroundings leads to a 'reversal' of the process. (The system is very close to equilibrium and infinitesimal changes can restore the system and surroundings to the original state).

□ If a block of material (at T) is in contact with surrounding at $(T - \Delta T)$, then 'heat will flow' into the surrounding. Now if the temperature of the surrounding is increased to $(T + \Delta T)$, then the direction of heat flow will be reversed.

□ If a block of material (at 40°C) is contact with surrounding at 80°C then the 'heat transfer' with takes place is not reversible.

□ Though the above example uses temperature differences to illustrate the point, the situation with other stimuli like pressure (differences) is also identical.

□ Consider a piston with gas in it a pressure ' P '. If the external pressure is $(P + \Delta P)$, then the gas (in the piston) will be compressed (slightly). The reverse process will occur if the external (surrounding pressure is slightly lower).

□ Maximum work will be done if the compression (or expansion) is carried out in a reversible manner.

Reversible process NOT a Reversible process

Heat flow Heat flow Heat flow

direction direction direction

T T 40°C

$T - \Delta T$

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$T + \Delta T$ 80°C 17

How to visualize a 'reversible' equivalent to a 'irreversible' processes?

□ Let us keep one example in mind as to how we can (sometimes) construct a 'reversible'

equivalent to a 'irreversible' processes.

□ Let us consider the example of the freezing of 'undercooled water'* at -5°C (at 1 atm pressure). This freezing of undercooled water is irreversible (P1 below).

□ We can visualize this process as taking place in three reversible steps □ hence making the entire process reversible (P2 below).

Irreversible

P1 Water at -5°C Ice at -5°C

Water at -0°C Ice at 0°C

P2 Heat Reversible Cool

* 'Undercooled' implies that the water Water at -5°C Ice at -5°C

is held in the liquid state below

the bulk freezing point! How is

this possible?→ read chapter on

phase transformations

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Reversible P-V work on a closed system

□ In a closed system (piston in the example figure below), if infinitesimal pressure increase causes the volume to decrease by ΔV , then the work done on the system is:

□ The system is close to equilibrium during the whole process dwreversible □ □ PdV thus making the process reversible.

□ As ΔV is negative, while the work done is positive (work done on the system is positive, work done by the system is negative).

If the piston moves outward under influence of P (i.e. 'P' and ΔV are in opposite directions, then work done is negative.

1

Note that the 'P' is the pressure inside the container. For the work to be done reversibly the pressure outside has to be $P+\Delta P$ ($\sim P$ for now). Since P the piston is moving in a direction opposite to the action of P, the work ($P+\Delta P$) done by the surrounding is $P\Delta V$ (or the work done by the system is $\Delta P\Delta V$, 2 i.e. negative work is done by the system).

□ 'Ultimately', all forms of energy will be converted to heat!!

□ One nice example given by Atkins: consider a current through a heating wire of a resistor. There is a net flow of electrons down the wire (in the direction of the potential gradient)

□ i.e. work is being done.

Now the electron collisions with various scattering centres leading to heating of the wire

□ i.e. work has been converted into heat.

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State functions in TD

□ A property which depends only on the current state of the system (as defined by T, P, V etc.)

is called a state function. This does not depend on the path used to reach a particular state.

□ Analogy: one is climbing a hill- the potential energy of the person is measured by the height of his CG from 'say' the ground level. If the person is at a height of 'h' (at point P), then his potential energy will be mgh , irrespective of the path used by the person to reach the height (paths C1 & C2 will give the same increase in potential energy of mgh - in figure below).

□ In TD this state function is the internal energy (U or E). (Every state of the system can be ascribed to a unique U).

□ Hence, the work needed to move a system from a state of lower internal energy ($=U_L$) to a state of higher internal energy (U_H) is $(U_H - U_L)$. $W = (U_H - U_L)$

□ The internal energy of an isolated system (which exchanges neither heat nor mass) is constant □ this is one formulation of the first law of TD.

□ A process for which the final and initial states are same is called a cyclic process. For a cyclic process change in a state function is zero.

E.g. $\Delta U(\text{cyclic process}) = 0$.

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Spontaneous and Driven processes

□ A spontaneous process is one which occurs 'naturally', 'down-hill' in energy*. I.e. the process does not require input of work in any form to take place.

□ Melting of ice at 50°C is a spontaneous process.

□ A driven process is one wherein an external agent takes the system uphill in energy (usually by doing work on the system).

□ Freezing of water at 50°C is a driven process (you need a refrigerator, wherein electric current does work on the system).

□ Later on we will note that the entropy of the universe will increase during a spontaneous change. (I.e. entropy can be used as a single parameter for characterizing spontaneity).

* The kind of 'energy' we are talking about depends on the

conditions. As in the topic on Equilibrium, at constant

temperature and pressure the relevant TD energy is Gibbs Spontaneous process

(Click to see)

free energy.

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Heat Capacity

□ Heat capacity is the amount of heat (measured in Joules or Calories) needed to raise an unit amount of substance (measured in grams or moles) by an unit in temperature (measured in $^\circ\text{C}$ or K). As mentioned before bodies (systems) contain internal energy and not heat.

□ This 'heating' (addition of energy) can be carried out at constant volume or constant pressure. At constant pressure, some of the heat supplied goes into doing work of

expansion and less is available with the system (to raise its temperature).

□ Heat capacity at constant Volume (CV): □ □E □

C □ □ □

It is the slope of the plot of internal energy with temperature. □ □T □V

V

□ Heat capacity at constant Pressure (CP):

□ □H □

It is the slope of the plot of enthalpy with temperature. CP □ □ □

□ Units: Joules/Kelvin/mole, J/K/mole, J/□C/mole, J/□C/g. □ □T □P

□ Heat capacity is an extensive property (depends on 'amount of matter')

□ If a substance has higher heat capacity, then more heat has to be added to raise its temperature. Water with a high heat capacity (of CP = 4186 J/K/mole = 1 Cal/□C/Kg) heats up slowly as compared to air (with a heat capacity, CP = 29.07J/K/mole) □ this implies that oceans will heat up slowly as compared to the atmosphere.

□ As T□0K, the heat capacity tends to zero. I.e near 0 Kelvin very little heat is required to raise the temperature of a sample. (This automatically implies that very little heat has to be added to raise the temperature of a material close to 0K.

This is of course bad news for cooling to very low temperatures□ small leakages of heat will lead to drastic increase in temperature).

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First Law of Thermodynamics

- The first law of thermodynamics is an extension of the law of conservation of energy
- The change in internal energy of a closed system will be equal to the energy added to the system minus the work done by the system on its surroundings

$$\Delta U = Q - W$$

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Process Terminology

- Adiabatic – no heat transferred
- Isothermal – constant temperature
- Isobaric – constant pressure
- Isochoric – constant volume

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Adiabatic Process

- An adiabatic process transfers no heat
– therefore $Q = 0$
- $\Delta U = Q - W$
- When a system expands adiabatically, W is positive (the system does work) so ΔU is negative.
- When a system compresses adiabatically, W is negative (work is done on the system) so ΔU is positive.

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Isothermal Process

- An isothermal process is a constant temperature process.
Any heat flow into or out of the system must be slow enough to maintain thermal equilibrium
- For ideal gases, if ΔT is zero,
 $\Delta U = 0$

- Therefore, $Q = W$
– Any energy entering the system (Q) must leave as work (W)

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Isobaric Process

- An isobaric process is a constant pressure process. ΔU , W , and Q are generally non-zero, but calculating the work done by an ideal gas is straightforward
 $W = P \cdot \Delta V$

- Water boiling in a saucepan is an example of an isobar process

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Isochoric or isovolumetric Process

- An isochoric process is a constant volume process.
When the volume of a

system doesn't change, it
will do no work on its
surroundings. $W = 0$

$$\Delta U = Q$$

- Heating gas in a closed container is an isochoric process

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Heat Capacity

- The amount of heat required to raise a certain mass of a material by a certain temperature is called heat capacity

$$Q = mc\Delta T$$

- The constant c_x is called the specific heat of substance x , (SI units of $\text{J/kg}\cdot\text{K}$)

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Heat Capacity of Ideal Gas

- C_V = heat capacity at constant volume

$$C_V = \frac{3}{2} R$$

- C_P = heat capacity at constant pressure

$$C_P = \frac{5}{2} R$$

- For constant volume

$$Q = nC_V\Delta T = \Delta U$$

- The universal gas constant $R = 8.314 \text{ J/mol}\cdot\text{K}$

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Example

5000 J of heat are added to two moles of an ideal monatomic gas, initially at a temperature of 500 K, while the gas performs 7500 J of work. What is the final temperature of the gas?

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Example

Compute the internal energy change and temperature change for the two processes involving 1 mole of an ideal monatomic gas.

- (a) 1500 J of heat are added to the gas and the gas does no work and no work is done on the gas
- (b) 1500 J of work are done on the gas and the gas does no work and no heat is added or taken away from the gas

Notice that in both processes, the change in internal energy is the same.

We say that the internal energy is a “state function”. A state function depends only on the state of the system and not on the process that brings the system to

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NEED FOR THE SECOND LAW

- The First Law of Thermodynamics tells us that during any process, energy must be conserved.
- However, the First Law tells us nothing about in which direction a process will proceed spontaneously.
- It would not contradict the First Law if a book suddenly jumped off the table and maintained itself at some height above the table.
- It would not contradict the First Law if all the oxygen molecules in the air in this room suddenly entered a gas cylinder and stayed there while the valve was open.

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The Second Law of Thermodynamics

The absence of the process illustrated above indicates that conservation of energy is not the whole story. If it were, movies run backwards would look perfectly normal to us!

The 2nd Law can also be stated that heat flows spontaneously from a hot object to a cold object

(spontaneously means without the assistance of external work)

Another statement of the second law of thermodynamics: The total entropy of an isolated system never decreases.

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MEANING OF ENTROPY AND THE SECOND LAW

- Entropy is a measure of the disorder (randomness) of a system. The higher the entropy of the system, the more disordered it is.
- The second law states that the universe always becomes more disordered in any real process.
- The entropy (order) of a system can decrease, but in order for this to happen, the entropy (disorder) of the surroundings must increase to a greater extent, so that the total entropy of the universe always increases.

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Concerning the 2nd Law

- The second law of thermodynamics introduces the notion of entropy (S), a measure of system disorder (messiness)
- U is the quantity of a system's energy, S is the quality of a system's energy.
- Another C.P. Snow expression:
 - not knowing the 2nd law of thermodynamics is the cultural equivalent to never having read Shakespeare

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Implications of the 2nd Law

- Time marches on
 - If you watch a movie, how do you know that you are seeing events in the order they occurred?
 - If I drop a raw egg on the floor, it becomes extremely “disordered” (greater Entropy) –

playing the movie in reverse would show pieces coming together to form a whole egg (decreasing Entropy) – highly unlikely!

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Direction of a Process

- The 2nd Law helps determine the preferred direction of a process
- A reversible process is one which can change state and then return to the original state
- This is an idealized condition – all real processes are irreversible

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Example

Suppose 0.1 kg ice at 0°C (273K) is in 0.5kg water at 20°C (293K) Calculate Heat transfers

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Example

Suppose 0.1 kg ice at 0°C (273K) is in 0.5kg water at 20°C (293K). What is the change in entropy of the ice as it melts at 0°C?

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Heat Engine

- It is easy to produce thermal energy using work, but how does one produce work using thermal energy?
 - This is a heat engine; mechanical energy can be obtained from thermal energy only when heat can flow from a higher temperature to a lower temperature.
 - A device which transforms heat into work is called a heat engine
 - This happens in a cyclic process
 - Heat engines require a hot reservoir to supply energy (QH) and a cold reservoir to take in the excess energy (QL)
- QH is defined as positive, QL is negative

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Heat Engines

A steam engine is one

- We will discuss only engines type of heat engine.

that run in a repeating cycle;

the change in internal energy

over a cycle is zero, as the

system returns to its initial

state.

- The high temperature

reservoir transfers an amount

of heat Q_H to the engine,

where part of it is transformed

into work W and the rest, Q_L ,

is exhausted to the lower

temperature reservoir. Note

that all three of these

quantities are positive.

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Heat Engines

The internal combustion engine is a type of heat

engine as well.

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Heat Engines

Why does a heat engine need a temperature difference?

Otherwise the work done on the system in one

part of the cycle will be equal to the work done

by the system in another part, and the net work

will be zero.

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Heat Engines

The efficiency of the heat engine is the ratio of

the work done to the heat input:

Using conservation of energy to eliminate W , we find:

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Heat Engines

The Carnot engine was created to examine the efficiency of a heat engine. It is idealized, as it has no friction. Each leg of its cycle is reversible.

The Carnot cycle consists of:

- Isothermal expansion
- Adiabatic expansion
- Isothermal compression
- Adiabatic compression

An example is on the next slide.

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Heat Engines

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Heat Engines

For an ideal reversible engine, the efficiency can be written in terms of the temperature:

From this we see that 100% efficiency can be achieved only if the cold reservoir is at absolute zero, which is impossible.

Real engines have some frictional losses; the best achieve 60-80% of the Carnot value of efficiency.

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3rd Law)

- No system can reach absolute zero
- This is one reason we use the Kelvin temperature scale. Not only is the internal energy proportional to temperature, but you never have to worry about dividing by zero in an equation!
- There is no formula associated with the 3rd Law of Thermodynamics

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Implications of 3rd Law

- MIT researchers achieved 450 picokelvin

in 2003 (less than $\frac{1}{2}$ of one billionth!)

- Molecules near these temperatures have been called the fifth state of matter:

Bose-Einstein Condensates

- Awesome things like super-fluidity and super-conductivity happen at these temperatures
- Exciting frontier of research

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The Zeroth Law

- The First and Second Laws were well entrenched when an additional Law was recognized (couldn't renumber the 1st and 2nd Laws)
- If objects A and B are each in thermal equilibrium with object C, then A and B are in thermal equilibrium with each other
- Allows us to define temperature relative to an established standard

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Temperature Standards

- See Heat versus Temperature slides for a discussion of these two concepts, and the misconceptions surrounding them
- Heat is energy transfer
- Temperature is proportional to internal energy
- Fahrenheit, Celsius, and Kelvin temp scales

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